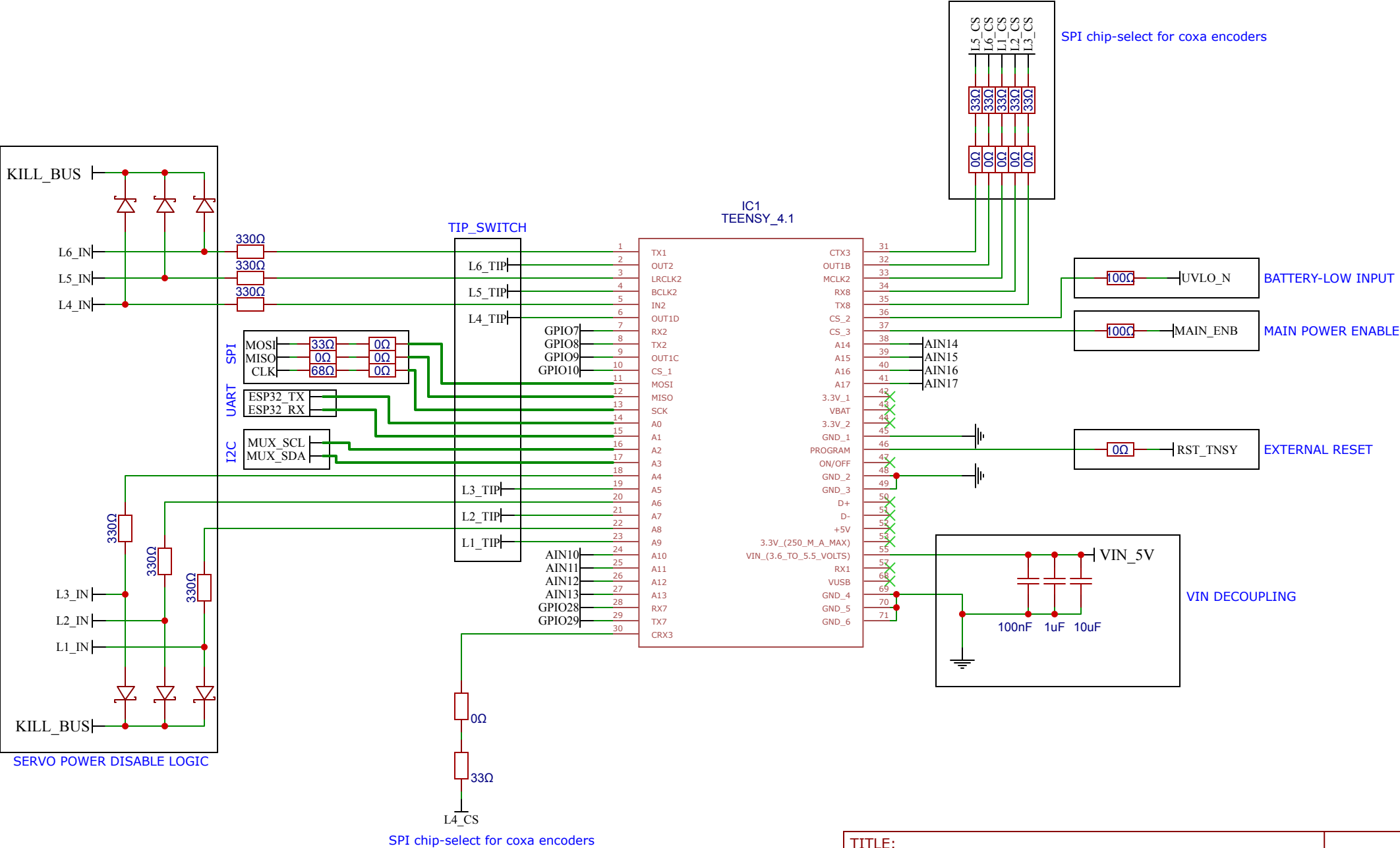


TEENSY 4.1 CONTROLLER



TITLE: MAIN_logic_tensy		REV: 1.0
EasyEDA	Company:	Sheet: 1/1
	Date: 2026-05-28	Drawn By: enricomariotto91

5V supply input from the main power board
1812L300/24SLER PTC populated by default
0468003.NRHF fuse holder footprint is optional / DNP



POWER CONTROL INTERFACE

TEENSY RESET INTERFACE

Components shown in parentheses are DNP by default.

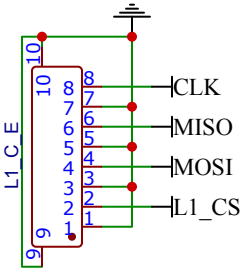
TITLE: MAIN_logic_SYSTEM MANAGEMENT		REV: 1.0
	Company: Your Company	Sheet: 1/1
	Date: 2026-06-02 Drawn By: enricomariotto91	

ENCODER INTERFACE

Components shown in parentheses are DNP by default.

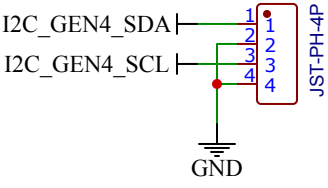
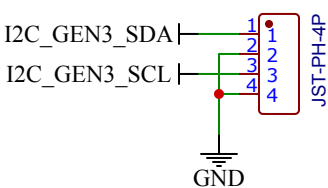
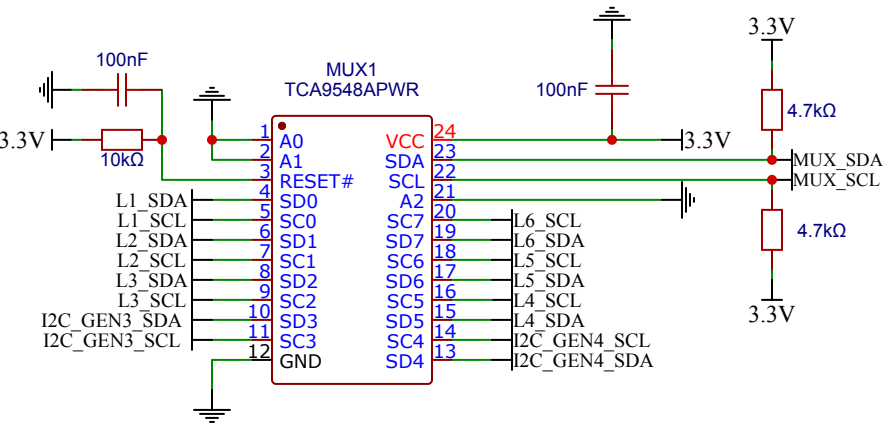
COXA ENCODER SPI INTERFACE

This interface is replicated for each leg
Conn type: JST-SH 1.0mm 8-PIN SMD



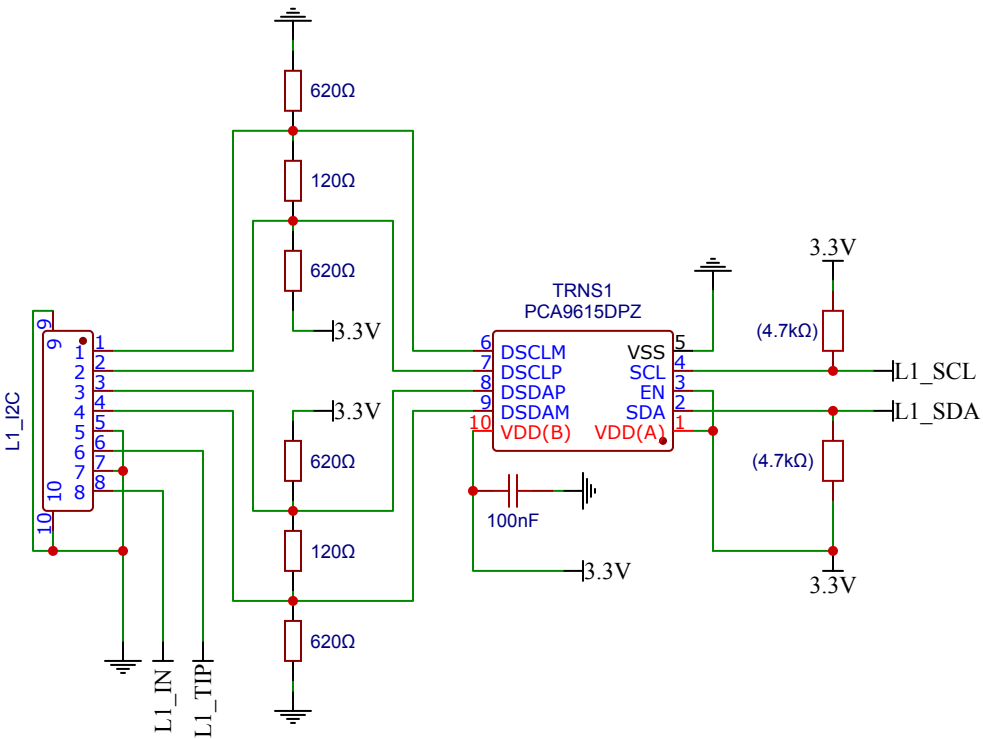
I2C MULTIPLEXER

Six channels are dedicated to leg modules and connect to differential I2C transceivers
Two additional channels are routed to auxiliary expansion connectors for future use



I2C DIFFERENTIAL INTERFACE

Repeated for all six leg communication channels
Conn type: JST-SH 1.0mm 8-PIN SMD



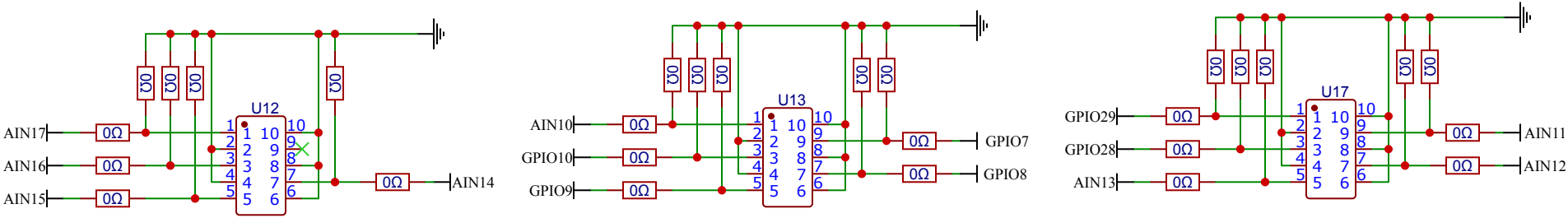
TITLE: MAIN_logic_encoder_interface		REV: 1.0
Company:		Sheet: 1/1
Date: 2026-05-28	Drawn By: enricomariotto91	

TEENSY GENERAL PURPOSE I/O

Unused Teensy GPIO and analog input signals are routed to expansion headers

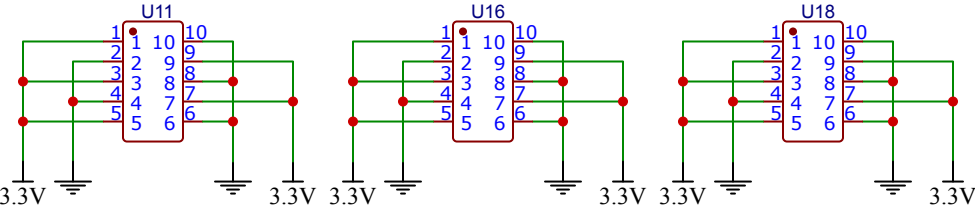
Optional series resistors and pull-down footprints are provided for signal conditioning and are not populated by default

Conn type: 2.54mm 10-PIN HEADER



3.3V AUXILIARY OUTPUTS

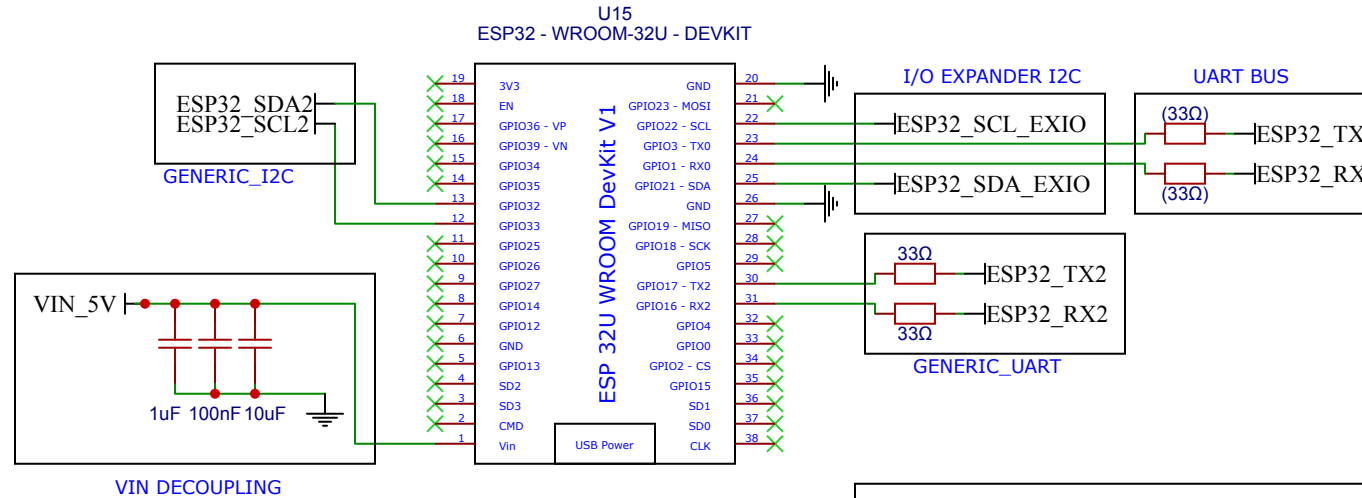
Conn type: 2.54mm 10-PIN HEADER



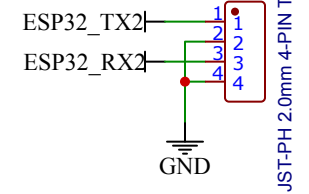
Components shown in parentheses are DNP by default.

TITLE: MAIN_logic_teensy_general_purpose_I/O		REV: 1.0
EasyEDA	Company: Your Company	Sheet: 1/1
	Date: 2026-06-02	Drawn By: enricomariotto91

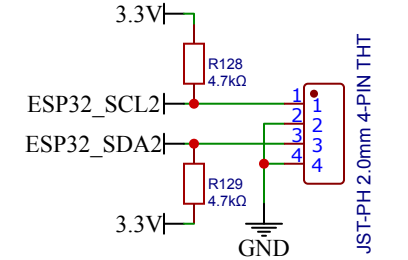
ESP32 COMMUNICATION & I/O



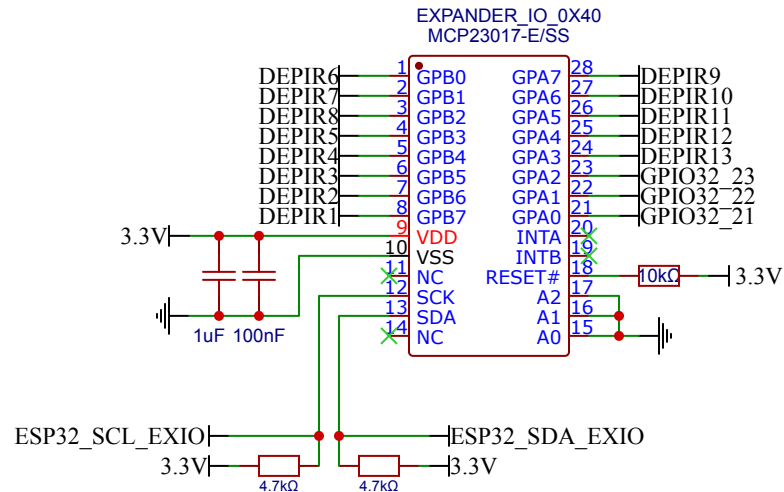
ESP32_UART2



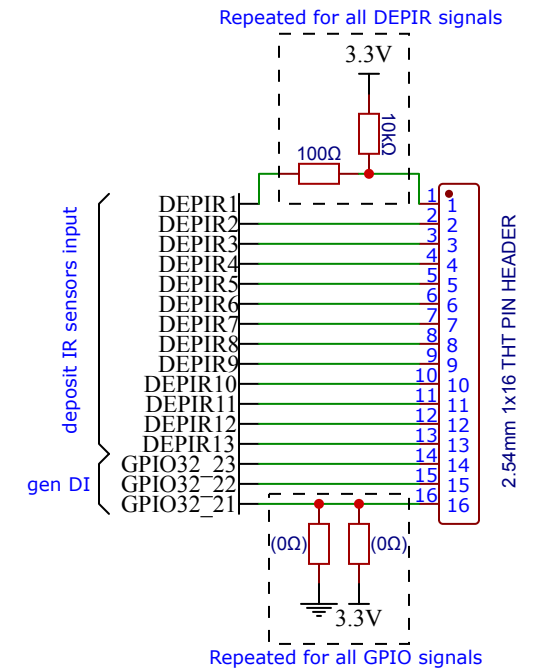
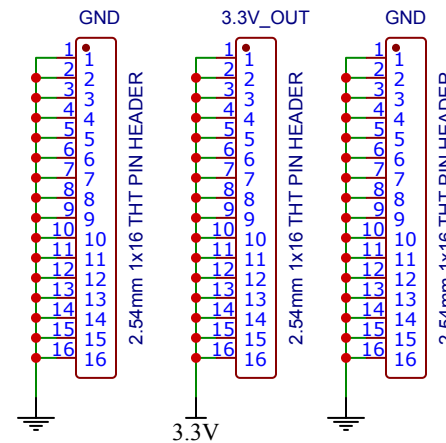
ESP32_I2C



AUXILIARY COMMUNICATION INTERFACES



DEPOSIT POSITION INTERFACES



Components shown in parentheses are DNP by default.

TITLE: ESP32		REV: 1.0
Company: Your Company		Sheet: 1/1
Date: 2026-05-28		Drawn By: enricomariotto91